

Bhupendra Kumar

bkumar0823@gmail.com | (+91) 8864866262 | <https://bkumar0823.space>

SUMMARY

Software Engineer with 2.5+ years of experience in designing and developing scalable backend systems, AI-powered applications, and full-stack web solutions. Experienced in Python, FastAPI, React.js, REST APIs, MongoDB, and Large Language Models (LLMs) with hands-on experience building Retrieval-Augmented Generation (RAG) systems, multi-tenant AI platforms, document processing pipelines, and enterprise web applications. Passionate about Generative AI, backend architecture, API development, and building production-ready AI applications.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, JWT Authentication, RBAC

Frontend: React.js, Angular, HTML5, CSS3, Bootstrap, MUI

Database: MongoDB, SQL, PostgreSQL

Generative AI: Langchain, Langgraph, LLMs, RAG, Prompt Engineering, Embedding, Vector Databases, Semantic Search, Ollama

Tools & Platforms: Git, GitHub, Postman, VS Code, AWS (Basic)

, Microsoft Azure

EXPERIENCE

System Engineer | Tata Consultancy Services (TCS) | New Delhi

December 2023 – Present

- Designed and developed enterprise AI applications using Python, FastAPI, React.js, PostgreSQL, REST APIs, and Large Language Models (LLMs).
- Contributing to an AI-powered healthcare Claim Audit platform, developing backend services and AI workflows to automate evidence collection and streamline audit processes.
- Developing Agentic AI workflows using LangGraph for intelligent task orchestration, evidence retrieval, and AI-assisted claim analysis.
- Building scalable FastAPI services integrating PostgreSQL databases, AI components, and enterprise REST APIs.
- Engineered configurable Retrieval-Augmented Generation (RAG) pipelines supporting document ingestion, chunking strategies, vector retrieval, prompt customization, and AI response generation.
- Developed a multi-tenant AI platform featuring Super Admin, Tenant Admin, and End User roles with secure tenant isolation and configurable AI workflows.
- Built scalable backend APIs for authentication, user management, document processing, AI inference, and knowledge retrieval.
- Implemented secure authentication and authorization using JWT and Role-Based Access Control (RBAC).
- Developed proof-of-concept AI applications including a Natural Language-to-Database Query Generator and an AI Resume Analyzer utilizing prompt engineering and semantic similarity techniques.
- Collaborated with cross-functional teams following Agile methodologies, participating in architecture discussions, API design, code reviews, testing, debugging, and deployment.
- Environment:** Python, FastAPI, React.js, PostgreSQL, MongoDB, LangGraph, Agentic AI, LLMs, RAG, Prompt Engineering, REST APIs, JWT, Git

Software Engineer | Leasing Monk | Noida

Nov 2022 – Feb 2023

- Completed full-stack web development training focused on modern JavaScript technologies.
- Developed responsive UI components using React.js based on Figma designs.
- Collaborated with the development team to build reusable frontend components and improve application usability.
- Worked with REST APIs and frontend integration following component-based architecture.
- Environment:** React.js, JavaScript, Node.js, Express.js, MongoDB, Git

PROJECTS

AI-Powered Claim Audit & Evidence Collection Platform

Technologies: Python, FastAPI, React.js, PostgreSQL, LangGraph, Agentic AI, LLMs, REST APIs

- Contributing to the development of an AI-powered healthcare claim auditing platform to streamline evidence collection and audit workflows.
- Building Agentic AI workflows using LangGraph to automate evidence gathering, claim analysis, and intelligent audit assistance.
- Developing backend services using FastAPI for claim processing, audit orchestration, and secure REST API integration.
- Implementing React.js interfaces for auditors to investigate claims, review evidence, and manage audit activities.
- Working with PostgreSQL to manage healthcare claim data, audit records, and application metadata.
- Developing AI-assisted capabilities to simplify claim analysis, reduce manual investigation effort, and improve audit efficiency.
- Collaborating within an agile team to design scalable, modular, and maintainable AI-driven healthcare solutions.

Multi-Tenant Enterprise RAG Platform

Technologies: FastAPI, React.js, Python, LLMs, RAG, Vector Database, JWT Authentication, REST APIs

- Designed and implemented a multi-tenant enterprise RAG platform supporting Super Admin, Tenant Admin, and End User roles.
- Built tenant onboarding and user management modules allowing organizations to independently manage AI configurations and knowledge repositories.
- Developed configurable RAG pipelines supporting document upload, chunking strategies, Top-K retrieval, temperature, and prompt customization.
- Implemented secure authentication, document management, vector search integration, and AI-powered question answering.
- Designed scalable backend architecture with tenant-level data isolation and configurable AI workflows.

Natural Language to Database Query Generator (Proof of Concept)

Technologies: Python, FastAPI, MongoDB, LLMs

- Developed an AI-assisted system to convert natural language questions into structured database queries using LLM reasoning.
- Implemented validation mechanisms for secure query generation before execution.
- Built REST APIs to process user requests and execute generated queries against backend databases.
- Explored prompt engineering techniques for accurate and safe query generation.

AI Resume Analyser & Job Matching Tool

Technologies: Python, LLMs, Prompt Engineering, Embedding

- Built an AI-assisted resume analysis system capable of extracting skills and generating improvement suggestions.
- Compared resumes with job descriptions using semantic similarity techniques.
- Designed structured prompts for consistent and well-formatted AI responses.
- Explored embedding-based approaches for intelligent resume-job matching.

Food Ordering Web Application

Technologies: Angular, Node.js, Express.js, MongoDB

- Developed a full-stack restaurant management and food ordering application.
- Implemented customer ordering workflows, kitchen management, authentication, and REST APIs.
- Designed a modular architecture to improve maintainability and scalability.

EDUCATION

Shri Ram Murti Smarak College of Engineering and Technology, Bareilly (*AKTU*) 2019 – 2023
B.Tech in CSE– Percentage: 82.0%.

Girish Prasad Memorial School, Bareilly (*ISCE Board*) 2018 – 2019
Intermediate – Percentage: 79.0%.

Girish Prasad Memorial School, Bareilly (*ISC Board*) 2016 – 2017
High School – Percentage: 81.83%.

CERTIFICATIONS

- Microsoft Azure Data Fundamentals (**DP-900**) – July 2024

CORE COMPETENCIES

- Generative AI Application Development
- FastAPI Backend Development
- REST API Design
- Retrieval-Augmented Generation (RAG)
- Prompt Engineering
- LLM Integration
- Authentication & Authorization
- Agile Development
- Git Version Control